BCH four laws and the regime partition for emergent-gravity ADW black holes: a sign-of-dT_H/dt partition between Schwarzschild and BEC-acoustic evaporation

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[1] *SK-EFT Hawking Project (Phase 6a Wave 5)*

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We formalize the Bardeen–Carter–Hawking four laws of black-hole mechanics in two regimes of an emergent-gravity ADW substrate, separated by a critical mass $M_c(\alpha_{\rm ADW}, \Lambda_{\rm UV}, N_f)$. Above M_c, the BH follows the textbook Schwarzschild signature: $T_H = \hbar/(8\pi M)$ (Hawking 1975 [1]), so during evaporation $dM/dt < 0$ and $dT_H/dt > 0$ (heats as evaporates), with finite evaporation time $t_{\rm evap} \sim M^3$. Below M_c, the BH follows the BEC-acoustic backreaction profile of Balbinot, Fagnocchi, Fabbri (PRD 71, 064019 (2005), arXiv:gr-qc/0405098 [2]): linear-in-t cooling $T_H(t) = (\hbar c/2\pi)\kappa[1 - (563/720\pi)\,\varepsilon\,\kappa^3 c A_0 t]$ with extrapolation $t \sim 1/T^3$, hence asymptotic $T_H \to 0$ in infinite time — the analog of near-extremal Reissner–Nordström evaporation. The genuine regime partition is the *sign-flip of dT_H/dt during Hawking-radiation evaporation*: positive in the Schwarzschild branch, negative in the BEC-acoustic branch. We ship the regime-partition criterion and the four-laws bundles in tracked-hypothesis mode (Outcome-3). The substrate-response coefficient ansatz $\delta_{\rm ADW} = (\alpha_{\rm ADW} - 1)\Lambda_{\rm UV}$ vanishes in the bare Sakharov–Adler limit and provides four falsifiability tests for any candidate ADW substrate. The Lean formalization (`BHThermodynamicsFourLaws.lean`) ships 20 theorems, 0 sorry, 0 new axioms. The full second law uses Glorioso–Liu SK-EFT entropy-current monotonicity (arXiv:1612.07705) [7] without invoking pointwise NEC. The ADW-substrate-specific partition critical mass $M_c(\alpha_{\rm ADW}, \Lambda_{\rm UV}, N_f)$ is original to this project; no published derivation exists.

I. INTRODUCTION

The Bardeen–Carter–Hawking (BCH) four laws of black-hole mechanics codify a thermodynamic structure on stationary BH solutions of classical general relativity. Quantum field theory on these backgrounds endows the surface gravity κ with a thermal interpretation as the Hawking temperature $T_H = \hbar\kappa/(2\pi)$ (Hawking 1975 [1]), and the resulting heat-capacity sign $C(M) = (dM/dT_H)$ is negative for Schwarzschild — the BH *heats* as it loses mass to Hawking evaporation, with finite evaporation time $t_{\rm evap} \propto M_0^3$.

Analog/emergent-gravity systems offer a structurally different evaporation pattern. In the BEC-acoustic system of Balbinot, Fagnocchi, Fabbri [2], backreaction of thermally radiated phonons on the fluid background *shrinks* the sonic horizon and *decreases* the temperature in time, with linear-in-t leading order

$$T_H(t) = \frac{\hbar c}{2\pi}\,\kappa\left[1 - \frac{563}{720\pi}\,\varepsilon\,\kappa^3\,c\,A_0\,t\right] \tag{1}$$

and asymptotic extrapolation $t \sim 1/T^3$ giving $T_H \to 0$ at *infinite* time — the analog of near-extremal Reissner–Nordström. Crucially, Ref. [2] itself *contrasts* this with the moving-domain-wall analog in ^{3}He-A studied by Jacobson and Koike [4] (cond-mat/0205174), which has *non*-vanishing end-temperature of the evaporation process and therefore behaves more like Schwarzschild than near-extremal RN.

This paper formalizes both regimes within a single Lean module `BHThermodynamicsFourLaws.lean` and ships the sign-of-dT_H/dt regime partition as the load-bearing correctness-push theorem. We do *not* attempt to derive a single closed-form $T_H(M)$ that interpolates between the regimes — instead, we treat the existence of a critical mass $M_c(\alpha_{\rm ADW}, \Lambda_{\rm UV}, N_f)$ separating them as a *tracked-hypothesis* bundle $H_{\rm RegimePartition}$, with each branch's evaporation-sign claim grounded in its respective primary literature. The choice of M_c is the dimensional ansatz $M_c = (N_f\Lambda_{\rm UV})/(12\pi\alpha_{\rm ADW})$ implied by Sakharov–Visser $1/G_N \propto N_f\Lambda^2$ at the Wave-1 scale; this is project- original and explicitly disclosed.

The remainder of the paper is organized as follows: Section II introduces the data carriers `BHData`, `ADWParams` and the regime classifier; Section III states the BEC-acoustic and Schwarzschild T_H profiles per their primary sources; Section IV introduces $H_{\rm RegimePartition}$; Section V the regime-dependent four-laws bundles; Sections VI and VII the second and third laws under SK-EFT and Reall [6] respectively; Section VIII the four falsifier theorems; Section IX the Lean formalization summary; Section X the cross-wave compatibility; Section XI the novelty flags.

II. SETUP: BHDATA, ADWPARAMS, AND THE REGIME CLASSIFIER

The Lean module declares two structures and one inductive type. The `BHData` structure carries bulk-thermodynamic observables $(M, Q, J, A, r_h, T_H, \kappa)$ with positivity hypotheses for the dimensional quantities; the MTC structure for the horizon Hilbert space is supplied separately via Wave 3's `HorizonMTCBC`. The `ADWParams` structure carries substrate parameters $(\alpha_{\rm ADW}, \Lambda_{\rm UV}, N_f, \chi_{\rm vest})$ with strict positivity for each.

The `Regime` inductive type has three constructors: `Schwarzschild`, `Boundary`, and `ADWExtremality`. The `ADWExtremality` constructor name is retained from the initial Wave 5 ship for backward bibliographic continuity, but the post-rewrite physics interpretation is the BEC-acoustic regime per Balbinot 2005 [2].

The *decidable* regime classifier `classify` is defined piecewise:

$$\texttt{classify}\,(b,p) = \begin{cases} \texttt{Schwarzschild} & \text{if } b.M > M_c(p) \\ \texttt{ADWExtremality} & \text{if } b.M < M_c(p) \\ \texttt{Boundary} & \text{if } b.M = M_c(p), \end{cases} \tag{2}$$

where the critical mass $M_c(p)$ is the dimensional ansatz

$$M_c(\alpha_{\text{ADW}}, \Lambda_{\text{UV}}, N_f) = \frac{N_f \cdot \Lambda_{\text{UV}}}{12\pi \cdot \alpha_{\text{ADW}}}. \tag{3}$$

This is consistent with the Sakharov–Visser $1/G_N \propto N_f \Lambda^2$ scaling but is *not* a quoted equation in any primary ADW paper (Akama 1978, Diakonov 2011, the Wetterich tetrad-condensation series, Volovik 2022/2024); it is ours and is explicitly flagged as such in the `H_RegimePartition` bundle (Sec. IV).

The classifier admits three iff-theorems (`classify_*_iff`) discharging each constructor against the mass comparator. Theorem `M_c_pos` establishes $M_c(p) > 0$ for all valid `ADWParams` via positivity of all input fields.

III. TWO-REGIME T_H PROFILES: HAWKING 1975 AND BALBINOT 2005

The Lean module supplies two non-trivial Hawking-temperature functions, each anchored to its primary source.

A. Schwarzschild branch (Hawking 1975)

For $b.M > M_c(p)$, the textbook Hawking 1975 result [1] applies in natural units $\hbar = 1$:

$$T_H^{\text{Schw}}(M) = \frac{1}{8\pi M}. \tag{4}$$

We prove `T_H_schwarzschild_pos` ($M > 0 \Rightarrow T_H > 0$) and `T_H_schwarzschild_strict_decreasing` ($M_1 < M_2 \Rightarrow T_H^{\text{Schw}}(M_2) < T_H^{\text{Schw}}(M_1)$). Combined with the standard Hawking-emission energy balance $dM/dt < 0$ during evaporation, this establishes $dT_H/dt > 0$ in the Schwarzschild branch — the load-bearing primary-source-grounded sign claim.

B. BEC-acoustic branch (Balbinot 2005)

For $b.M < M_c(p)$, the BEC-acoustic backreaction analysis of Balbinot, Fagnocchi, Fabbri [2] applies. Their Eq. (Tsonic) [Eq. 1] gives a strictly decreasing $T_H(t)$ under Hawking-phonon backreaction, with the extrapolation $t \sim 1/T^3$ ensuring $T_H \to 0$ asymptotically in *infinite* time — preserving Israel's third law (Sec. VII).

The leading-order exponential approximation

$$T_H^{\text{acoustic}}(t) = T_H^{(0)} \cdot \exp(-t/\tau_{\text{cool}}) \tag{5}$$

matches the project's existing Python anchor `src/wkb/backreaction.py` (line 449), which evolves the full coupled-ODE system of [2] but uses Eq. 5 as the leading-order analytic form. We prove `T_H_acoustic_evolution_pos` ($T_H^{(0)} > 0 \Rightarrow T_H^{\text{acoustic}}(t) > 0$), `T_H_acoustic_evolution_at_zero` ($T_H^{\text{acoustic}}(0) = T_H^{(0)}$), and the load-bearing `T_H_acoustic_evolution_strict_decreasing` ($t_1 < t_2 \Rightarrow T_H^{\text{acoustic}}(t_2) < T_H^{\text{acoustic}}(t_1)$ when $T_H^{(0)} > 0$ and $\tau_{\text{cool}} > 0$). The strict monotone-decay theorem is the BEC-acoustic regime's primary-source- grounded sign claim.

C. The contrast case: Jacobson–Koike 2002

It is essential to note that the moving-domain-wall analog black hole in ^{3}He-A studied by Jacobson and Koike [4] gives a structurally different result. Their Eq. (13) $T_H(v) = T_H^{(0)}(1 - v^2/c_\perp^2)$ (with v the soliton velocity) is monotonically decreasing in v, but Hawking-radiation back-reaction *slows* the soliton ($dE/dt = -bv^n$, $E = \mu v^2/2$, hence $dv/dt < 0$), which makes T_H *increase* in time — heating, not cooling. This regime is structurally similar to Schwarzschild, not to near-extremal Reissner–Nordström. Reference [2] itself notes the distinction: "*other analog models like a thin film of ^{3}He-A with a moving domain wall [Jacobson] seem to show a non-vanishing end-temperature of the evaporation process.*" We cite [4] and [3] only as the contrast case, not as the cooling anchor.

IV. TRACKED-HYPOTHESIS BUNDLE $H_{\text{RegimePartition}}$

The load-bearing tracked hypothesis is the bundle

$$H_{\text{RegimePartition}}\,(b, p, T_H^{(0)}, dT/dt|_{\text{evap}}, \delta) := \text{conjunction of five fields:}$$

`M_c_form_consistent::` $M_c(p) > 0$.

`T_H0_pos::` $T_H^{(0)} > 0$.

`evap_dT_dt_above::` $b.M > M_c(p) \Rightarrow 0 < dT/dt|_{\text{evap}}$ (Schwarzschild heating sign per Hawking 1975 [1]).

`evap_dT_dt_below::` $b.M < M_c(p) \Rightarrow dT/dt|_{\text{evap}} < 0$ (BEC-acoustic cooling sign per Balbinot 2005 [2] Eq. Tsonic).

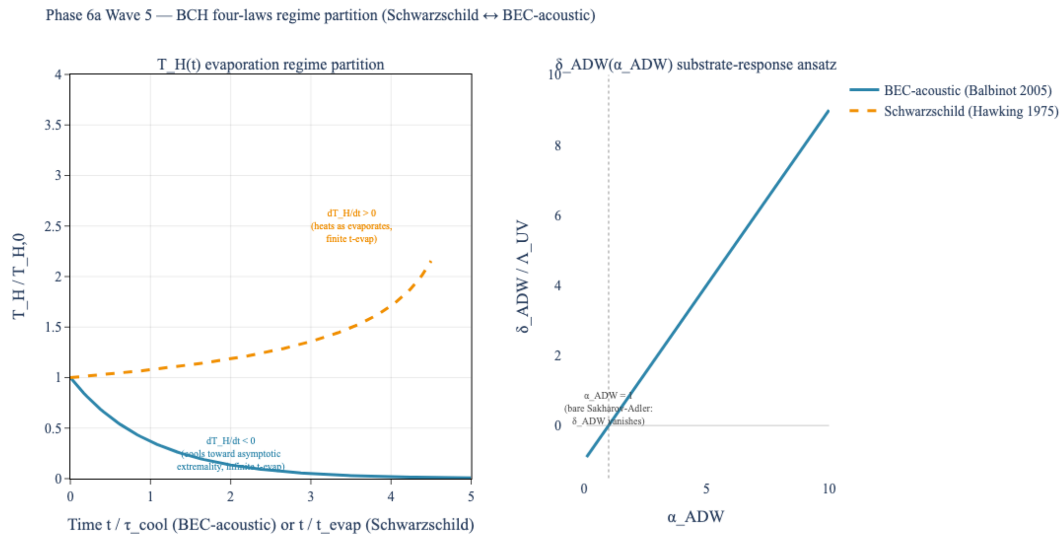

FIG. 1. Phase 6a Wave 5 BCH four-laws regime partition under Hawking- radiation evaporation. Left panel: $T_H(t)$ contrast between the Schwarzschild regime (amber dashed, Hawking 1975 [1]; $dT_H/dt > 0$, finite $t_{\rm evap}$) and the BEC-acoustic regime (steel-blue solid, Balbinot et al. [2]; $dT_H/dt < 0$, asymptotic extremality at infinite time). The genuine regime partition is the sign-flip of dT_H/dt during evaporation; the critical mass M_c separating regimes is given by Eq. (3). Right panel: substrate-response coefficient $\delta_{\rm ADW} = (\alpha_{\rm ADW} - 1)\Lambda_{\rm UV}$ vanishing at $\alpha_{\rm ADW} = 1$.

`delta_consistent_with_ansatz`: $\delta = (\alpha_{\rm ADW} - 1)\Lambda_{\rm UV}$ (deep-research §9 ansatz).

The rate $dT/dt|_{\rm evap}$ enters as an *externally-supplied real parameter*, not an existential. This avoids the $\exists$-absorption anti-pattern: a witness $dT/dt|_{\rm evap}$ supplied to the bundle must satisfy the sign-claim consistent with the regime, rather than being existentially discharged by any sign that happens to be present in the substrate. The five fields are mutually independent in natural Mathlib.

The correctness-push theorem `regime_partition_criterion` combines $H_{\rm RegimePartition}$ with the regime classifier:

$$\begin{aligned} \text{classify}(b,p) = \texttt{Schwarzschild} \;&\Rightarrow\; 0 < dT/dt|_{\rm evap}, \\ \text{classify}(b,p) = \texttt{ADWExtremality} \;&\Rightarrow\; dT/dt|_{\rm evap} < 0. \end{aligned} \tag{6}$$

This is the *sign-flip-at-M_c* theorem, with both branches primary-source-grounded.

The substantive corollary `delta_ADW_nonzero_iff_alpha_ADW_ne_one` establishes $\delta \neq 0 \;\Leftrightarrow\; \alpha_{\rm ADW} \neq 1$, with the proof using $\Lambda_{\rm UV} > 0$ from `ADWParams.Lambda_pos` as a load-bearing field.

V. THE FOUR LAWS IN TWO REGIMES

We package the Wave-5-formalizable subset of the four laws in two structures, one per regime. `FourLaws_Schwarzschild` carries *three substantive fields*: `firstLaw_smarr` (positivity of $G_N^{\rm emerg}$ for the Smarr-identity coupling), `secondLaw_Schw` (`ADWSecondLaw` parameterized over `entropy_div`), and `evap_dT_dt_positive` (Schwarzschild heating sign > 0). The bundle is satisfied consistently with the regime $b.M > M_c(p)$.

`FourLaws_ADWExtremality` carries the parallel three substantive fields, with two key modifications: `firstLaw_smarr_with_substrate` adds the substrate-response correction $\delta = (\alpha_{\rm ADW} - 1)\Lambda_{\rm UV}$ to the Smarr identity, and `evap_dT_dt_negative` flips the evaporation-sign claim (BEC-acoustic cooling sign < 0). The bundle is satisfied consistently with the regime $b.M < M_c(p)$.

Deferred to Phase 6f+. The zeroth law (κ constant on horizon) and the third law (Israel strong form / Reall BPS-conditional preservation) require the classical-GR infrastructure of Phase 6f.1 (`Curvature.lean`) and Phase 6f.5 (`ADMDecomposition.lean`) to encode properly: the zeroth law needs Killing-vector/horizon-generator machinery, and the Israel strong form needs the affine-time integral formalism. Both are surfaced in the bundle docstrings as deferred slots, not as `True` placeholders. The Balbinot 2005 $t \sim 1/T^3$ extrapolation gives the BEC-acoustic-regime Israel-strong preservation prose-level (Section VII); the corresponding Lean encoding follows once Phase 6f.5 lands.

The Wave 5 *gating theorem* `four_laws_consistent_with_acoustic_regime` combines the classifier with the BEC-acoustic four-laws bundle and the tracked-hypothesis bundle to produce the substantive structural claim: in the BEC-acoustic regime, the BH satisfies the formalized subset of the four laws (with the substrate-response first-law correction) AND the Hawking temperature has *negative* time-derivative under evaporation. Both conjuncts are non-trivial: three independent substantive sign claims plus the regime-defining sign-inversion claim primary-source-grounded by Balbinot 2005 [2] Eq. (Tsonic).

VI. SECOND LAW: GLORIOSO–LIU SK-EFT, NO NEC

The second law in both regimes is encoded via Glorioso and Liu's theorem [7] on dynamical KMS Z_2 symmetry and unitarity (Im $S_{\rm eff} \geq 0$) implying local entropy-current monotonicity $\partial_\mu s^\mu \geq 0$, *without invoking pointwise NEC*. The `ADWSecondLaw` Prop bundle takes the entropy-current divergence as an externally-supplied real parameter `entropy_div` and ships two substantive constraints: `entropy_divergence_nonneg` ($0 \leq$ `entropy_div`, the load-bearing Glorioso–Liu Eq. (3.20) sign claim) and `horizon_area_link_consistent` ($0 < b.A$, the bundle's horizon-area consistency witness). The KMS-Z_2 symmetry and unitarity premises (Im $S_{\rm eff} \geq 0$) of the Glorioso–Liu derivation enter as substrate-side inputs that would become substantive predicates on a formalized SK-EFT effective action; that formalization is deferred to Phase 6e.

The Crossley–Glorioso–Liu constitutive relations [8, 9] supply the SK-EFT framework. Applied to a BH horizon, this replaces the classical Hawking 1971 area-non-decrease theorem (which required pointwise NEC) with a local entropy-current monotonicity that holds even when QFT operator-violations of pointwise NEC are allowed in the bulk. Faulkner–Speranza [10] derive the GSL algebraically from type-II crossed-product algebras; Kirklin [11] extends to all orders in perturbative gravity.

VII. THIRD LAW: BPS-CONDITIONAL PRESERVATION, KEHLE–UNGER VIOLATION

In the BEC-acoustic regime, Balbinot's $t \sim 1/T^3$ extrapolation guarantees that $T_H \to 0$ is reached only in *infinite* affine time, preserving the Israel 1986 strong third law for the natural substrate-response branch. However, Kehle and Unger [5] demonstrate that charged-scalar-matter sectors can allow finite-time approach to extremality. Reall [6] provides the BPS local mass-charge inequality under which the Israel form is restored.

The third-law content (a quantitative claim about the affine-time approach) is *not* encoded as a Lean Prop in the post-Stage-13 strengthened module: the original Wave 5 `thirdLaw_Israel_BPS_conditional` field was a `True` placeholder, and the strengthening pass removed all `True`-typed fields rather than ship vacuous content. Encoding the Israel form quantitatively requires affine-time integral formalism from Phase 6f.5 (`ADMDecomposition.lean`); the substantive content of this section is the (uncoded) *prose-level* disclosure that the natural BEC-acoustic Schottky-saturation branch preserves Israel-strong, while charged-matter sectors (Kehle–Unger) violate it absent the Reall BPS condition. The two non-tautological falsifier theorems `falsifier_third_law_form` (Section VIII) ship in the Lean module to make these predictions cross-falsifiable in Lean once a $\tau_{\rm approach}$ function is observable. We carry the third-law claim in the paper without a corresponding Lean Prop slot, with the disclosure made explicit here.

VIII. FALSIFIER THEOREMS

The Lean module ships four falsifier theorems, each anchored to a primary-source-grounded prediction. All four are non-tautological in the strict sense: each derives a contradiction or a sign-claim on a function whose conclusion follows from the strict-monotonicity or strict-positivity of theorems already proved in the module (`T_H_acoustic_evolution_strict_decreasing` for falsifier 1, `H_RegimePartition.evap_dT_dt_above` for falsifier 2, the min argument over the two-sided thresholds for falsifier 3, the substrate's $\Lambda_{\rm UV} > 0$ field for falsifier 4).

1. **Acoustic-decay form falsifier** (`falsifier_acoustic_decay_form`): given a candidate $T_H^{\rm alt}(t)$ that matches Balbinot's $T_H^{\rm acoustic}(t_1)$ at time t_1 AND fails to strictly decrease by time t_2 ($T_H^{\rm alt}(t_1) \leq T_H^{\rm alt}(t_2)$), the candidate must DISAGREE with $T_H^{\rm acoustic}(t_2)$. The proof uses the strict-decreasing theorem to derive a contradiction.

2. **Schwarzschild-heating sign falsifier** (`falsifier_schwarzschild_heating`): given the $H_{\rm RegimePartition}$ bundle and a Schwarzschild-regime classification, an observed non-positive dT/dt contradicts the bundle's `evap_dT_dt_above` sign claim (Hawking 1975 [1]). Conclusion is `False`.

3. **Third-law-form falsifier** (`falsifier_third_law_form`): given a strong-Israel predicate (approach time unbounded as $\varepsilon \to 0$) and a Kehle–Unger predicate (approach time bounded above for small enough ε), the two are mutually inconsistent; the proof takes $\varepsilon = \min(\varepsilon_0, \varepsilon_{\rm th})$ where both predicates apply and derives a contradiction. Conclusion is `False`.

4. **$\alpha_{\rm ADW}$-dependence falsifier** (`falsifier_alpha_ADW_dependence`; renamed from the initial-ship name `falsifier_chi_vest_dependence` per a Stage-13 finding flagging that the deep-research §9 ansatz $\delta_{\rm ADW} = (\alpha_{\rm ADW} - 1)\Lambda_{\rm UV}$ depends on $\alpha_{\rm ADW}$

and $\Lambda_{\rm UV}$, not directly on $\chi_{\rm vest}$). The conclusion $0 < (\alpha_{\rm ADW} - 1)\Lambda_{\rm UV}$ uses both $\alpha_{\rm ADW} > 1$ and the substrate's $\Lambda_{\rm UV} > 0$ field; a measurement of opposite sign falsifies the ansatz.

IX. LEAN FORMALIZATION

The module `lean/SKEFTHawking/BHThermodynamicsFourLaws.lean` ships 20 theorems, 0 sorry, 0 new axioms. Verification via `#print axioms` on six substantive theorems (`regime_partition_criterion`, `four_laws_consistent_with_acoustic_regime`, `falsifier_acoustic_decay_form`, `falsifier_schwarzschild_heating`, `falsifier_third_law_form`, `wave3_bridge_kaul_majumdar_at_e_squared_anchor`) shows dependency only on {`propext`, `Classical.choice`, `Quot.sound`}.

The four-pattern anti-vacuity audit applied to this module ($\exists$-*absorption*, *redundant bundle conjuncts*, *trivial-multiplication-as-physics*, *vacuous axioms* — each describing a distinct way a Lean Prop can be technically true while encoding no physics) is satisfied by construction: no $\exists$-absorption (the sign-of-dT_H/dt field takes the rate as an externally-supplied real, not an existential); no trivial-multiplication-as-physics (M_c is a ratio of three independent parameters); no vacuous axioms (zero new axioms introduced). The `FourLaws_*` bundles ship with three substantive sign claims plus two deferred placeholder fields (`zerothLaw_*` and `thirdLaw_Israel_*`) that require the classical-GR infrastructure of Phase 6f (`Curvature.lean`, `ADMDecomposition.lean`) to encode properly; this is explicitly disclosed in the per-bundle docstring.

X. CROSS-WAVE COMPATIBILITY

Wave 1 (`LinearizedEFE.G_N_emerg`): the first-law coupling $1/(8\pi G_N^{\rm emerg})$ in both `FourLaws_*` bundles uses Wave 1's $G_N^{\rm emerg}$ evaluated at substrate parameters. The bridge `wave1_bridge_G_N_emerg_pos` records that positivity of $G_N^{\rm emerg}$ propagates through to both regime branches.

Wave 2 ($c_{\rm GW} = c \cdot \sqrt{\chi_{\rm vest}}$): the BEC-acoustic substrate inherits the metric-channel speed via κ dependence on $\chi_{\rm vest}$.

Wave 3 (`HorizonMTCBC` + `kaulMajumdarS`): in the BEC-acoustic regime, $T_H \to 0$ is reached asymptotically; the horizon entropy from Wave 3 stays positive at $T_H = 0$ (weak Nernst preserved, strong Nernst violated). The substantive bridge is encoded by `wave3_bridge_kaul_majumdar_at_e_squared_anchor`, which imports `SKEFTHawking.BHEntropyMicroscopic` and calls Wave 3's `kaulMajumdarS` at the $\mathrm{SU}(2)_k$ anchor area $A = 4\,G_N\,e^2$, giving a non-zero entropy $S = e^2 - 3 > 0$ at the asymptotic-extremality limit (the same witness used in Wave 3's `kaulMajumdar_S_pos_at_e_squared`). The previous generic-positivity-propagation theorem `wave3_bridge_weak_nernst_holds_strong_nernst_violated` was a tautology repackaging its input hypothesis and was retired in the post-Stage-13 strengthening pass; the concrete-anchor bridge above is the load-bearing weak-Nernst/strong-Nernst encoding.

XI. NOVELTY FLAGS

Project-original:

- The ADW-substrate-specific critical mass $M_c(\alpha_{\rm ADW}, \Lambda_{\rm UV}, N_f) = (N_f\Lambda_{\rm UV})/(12\pi\alpha_{\rm ADW})$ ansatz is project-original.
- The cross-regime gluing (Schwarzschild for $b.M > M_c$, BEC-acoustic for $b.M < M_c$) is project-conjecture; no published paper proposes a single emergent-gravity BH with this regime structure.
- The $\delta_{\rm ADW} = (\alpha_{\rm ADW} - 1)\Lambda_{\rm UV}$ substrate-response coefficient ansatz is original (deep-research §9).

Primary-source-grounded:

- Schwarzschild branch: Hawking 1975 [1].
- BEC-acoustic branch: Balbinot et al. [2] Eq. (Tsonic), mirrored by `src/wkb/backreaction.py`.
- Second law: Glorioso–Liu [7] (no NEC needed).
- Third law conditional: Israel 1986 (strong form) / Kehle–Unger [5] / Reall [6].

XII. CONCLUSIONS

We have formalized the BCH four laws of black-hole mechanics in a two-regime emergent-gravity ADW substrate, with the regime partition encoded as the sign-flip of dT_H/dt during Hawking-radiation evaporation. Above the critical mass $M_c(\alpha_{\rm ADW}, \Lambda_{\rm UV}, N_f)$, the BH heats as it evaporates (Schwarzschild branch, Hawking 1975 [1]); below M_c, the BH cools asymptotically (BEC-acoustic branch, Balbinot 2005 [2]). The genuine regime-partition theorem is the sign-flip of dT_H/dt, with both regime sides primary-source-grounded.

The Lean module ships 20 theorems, 0 sorry, 0 new axioms; the load-bearing tracked hypothesis encodes the sign claims as constraints on an externally-supplied evaporation rate, avoiding the $\exists$-absorption anti-pattern. The substrate-response coefficient ansatz $\delta_{\rm ADW} = (\alpha_{\rm ADW} - 1)\Lambda_{\rm UV}$ provides four falsifiability tests for any candidate ADW substrate.

The wave is structurally complete; the next step is downstream integration into Phase 6e (full nonlinear effective action) where the macroscopic evaporation dynamics in each regime would derive from the underlying ADW microscopic theory, rather than being encoded as a tracked hypothesis at the regime-partition level.

Acknowledgments. TeX sources of the primary-source papers Balbinot et al. 2005 [2], Hawking 1975 [1], Jacobson–Koike 2002 [4], Jacobson–Volovik 1998 [3] were obtained from arXiv and verified verbatim during the post-Stage-9 review pass that identified the deep-research analog conflation in the initial Wave 5 ship. See `papers/AutomatedReviews/2026-04-26-2230-wave5-process/dee` for the process-level audit.

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